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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester MCA (CBCS) Degree Examinations

Summer Semester July 2024

22MCA222 – HEALTH CARE ANALYTICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions.

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Discuss the concept of data and information in healthcare analytics and elaborate Describe the key measurements, metrics, and indicators used in healthcare analytics.	10	L*1	1	2
	b)	Elucidate the concepts of descriptive, predictive, and prescriptive analytics within the realm of healthcare, accompanied by illustrative examples showcasing their applications in the industry.	10	L1	1	2
2.	a)	List and describe the components involved in healthcare analytics. Support your explanation with relevant examples.	10	L2	1	2
	b)	Identify and describe different imaging modalities used in healthcare. Provide explanations for each modality.	10	L2	1	1
		Unit – II				
3.	a)	What is biomedical signal analytics, and why is it important in healthcare? Outline the difficulties related to analyzing sensor data in healthcare environments	10	L1	2	2
	b)	Describe the concept of cross-correlation analysis and elucidate the pre-processing steps involved in this analytical method.	10	L2	2	2
4.	a)	Explain the concept of bio signals in the time domain. Describe the different types of time-domain features commonly used for analyzing biomedical signals.	10	L1	2	1
	b)	Describe the typical processing pipeline for resting-state functional magnetic resonance imaging (rs-fMRI) processing, including a visual representation	10	L2	2	1
		Unit – III				
5.	a)	Explain the concept of clinicgenomic integration and examine the issues and challenges associated with integrating clinical and genomic data	10	L2	3	2
	b)	Describe the concept of dimensionality reduction, emphasizing its advantages and disadvantages	10	L1	3	2
6.	a)	Identify and describe the stages of data integration, highlighting the advantages and their disadvantages.	10	L1	3	2
	b)	What is genomic data? List and explain the different ways to generate genomic data.	10	L2	3	2
		Unit – IV				
7.	a)	Elaborate on the concept of Natural Language Processing (NLP) within biomedical information mining. Outline the fundamental components of NLP.	10	L2	4	1
	b)	Explore the obstacles encountered in processing clinical reports and provide examples of clinical applications associated with this process.	10	L2	4	1
8.	a)	Examine the utilization of social media analysis for research in public health.	10	L2	4	2

- b) Examine the significance of social media in analyzing, detecting, and tracking infectious disease outbreaks.

10 L2 4 1

Unit – V

9. a) Describe the different statistical prediction models employed in healthcare data analytics.
- b) Describe the differences between alternative clinical prediction models and statistical prediction models. Identify the strengths and weaknesses of each approach within healthcare data analytics.
10. a) Define temporal data mining within healthcare data analytics. Explore how association analysis and temporal pattern mining techniques are applied to derive insights from temporal healthcare data
- b) Discuss various alternative prediction models in healthcare data analytics, focusing on: i) Decision trees ii) Artificial neural networks iii) Cost-sensitive learning.

10 L1 5 2

10 L1 5 1

10 L1 5 1

10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
